

Office hours

- Fridays 12-1pm (zoom or Stewart Biology, N5/8)
<https://mcgill.zoom.us/j/81407275167?pwd=8Cgm0RPnWQRlOdaCNFyXnS3M8amQUj.1>
- By appointment
- Before or after class, right here in Leacock 132

Reminder: Self-Assessment Exercise on Wednesday March 12

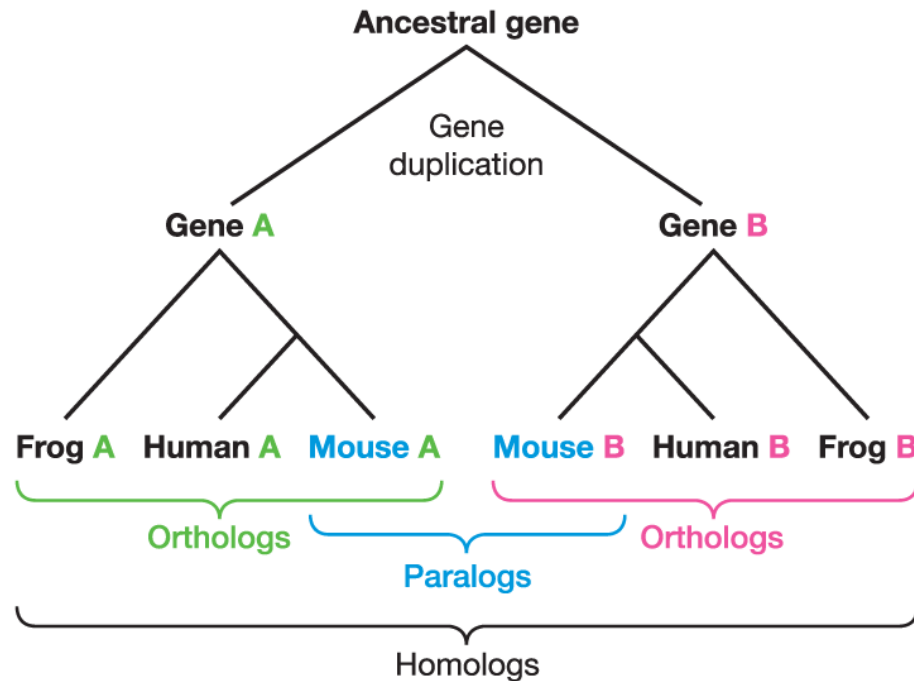
- 8 True/False Questions
- 8:35 am: lecture wrap-up
- 8:55 am: SAE

Sequence comparisons between species

Phylogenetic Inference

Comparing genomes between species: Phylogenetic Inference

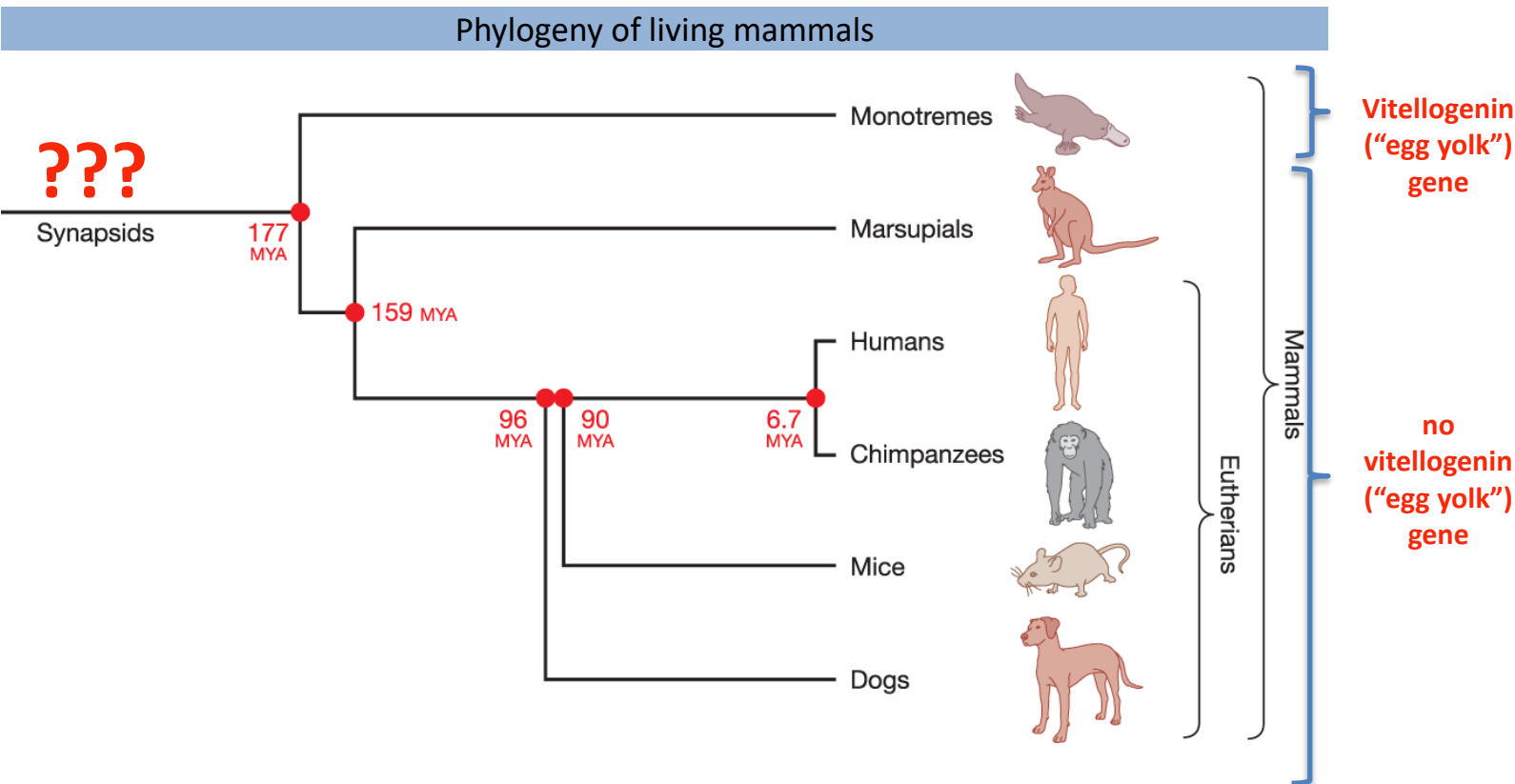
Relationships between homologs, orthologs, and paralogs



Orthologs: homologous genes at the same genetic locus in different species, evolutionarily inherited from a common ancestor

Paralogs: homologous genes at different loci in the same species, having arisen from gene duplication

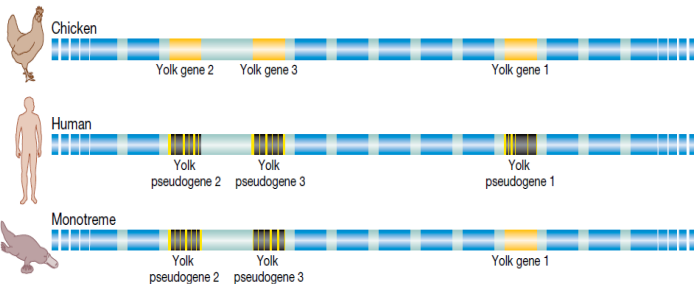
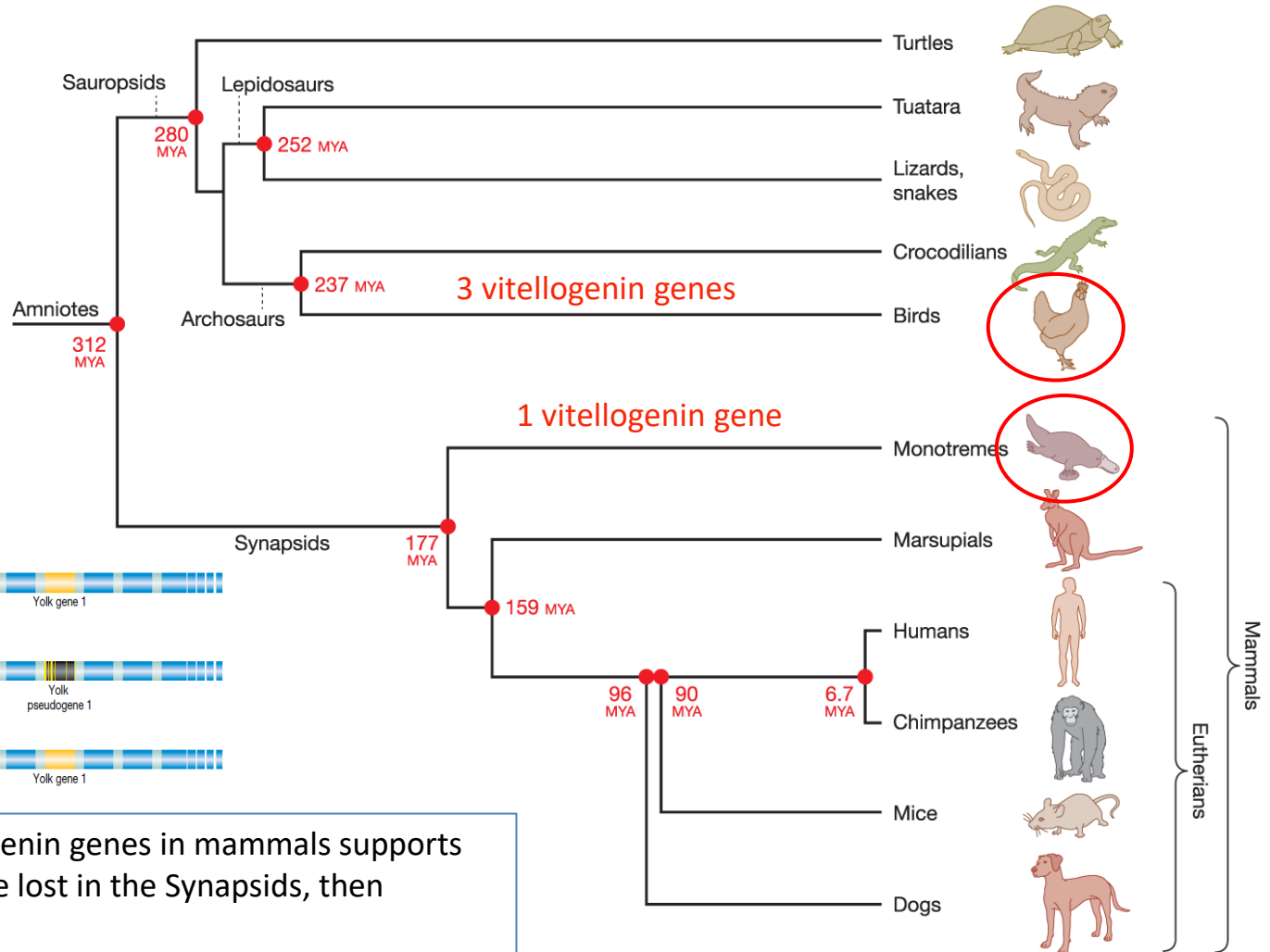
Example: using phylogenetic inference to understand how the platypus acquired its vitellogenin (egg yolk) gene.



The presence of the vitellogenin gene in the genome correlates with whether or not the animal lays eggs. Did monotremes GAIN the gene, or did the eutherians LOSE the gene? How do we tell? Ask if it was present in their last common ancestor.

Yolk genes were present in the common ancestor of mammals and birds.

Phylogeny of living mammals and other amniotes



Sequence comparisons within a single species

Sequence comparison within a genome can reveal families of related genes.

- Related genes within an organism that encode proteins of similar amino acid sequence
- Can contain 2 to >100 members (called **PARALOGS**)
- Members can be functionally redundant or have independent functions.
- Human keratin-associated protein 1 family. These proteins are involved in the structure of human hair fibers, and this family is **thought to have arisen through gene duplication during evolution.**

Deletion and duplication



human keratin-associated protein 1 family

hKAP1.1A/B	MACCQTSFCGFPSFSTGGTCDSSCCQPSCCETSSCCQPRCCETSSCCQPSCCQTS
hKAP1.2	MTCCQTSFCGYPSGSTSGTCGSSCCQPSCCETSSCCQPSCCETSSCCQPSCCQTS
hKAP1.3	MTCCQTSFCGYPSGSTSGTCGSSCCQPSCCETSSCCQPSCCETSSCCQPSCCQTS
hKAP1.4	MA-----SCSTSGTCGSSCCQPSCCETSSCCQPS-----
hKAP1.5	MTCCQTSFCGYPSFSGTCGSSCCQPSCCETSSCCQPS-----
hKAP1.6	MACCQTS-----
hKAP1.7	MACCQTSFCGFPSGST-----

N-terminal domain

hKAP1.1A/B	FCGFPSFSTGGTCDSSCCQPSCCETSSCCQPSCYQT-----SSCGTGCG
hKAP1.2	FCDFLASQLV-DLQLSCCQPSCCETSSCCQPSCCQT-----SSCGTGCG
hKAP1.3	FCGFPSFSTSGTCSSSCCQPSCCETSSCCQPSCCQT-----SSCGTGCG
hKAP1.4	-----CCQT-----SSCGTGCG
hKAP1.5	FCGFPSFSTSGTCSSSCCQPSCCETSSCCQPSCCETSSCCQPSCCQTS-----SSCGTGCG
hKAP1.6	FCGFRSFSTGGTCDSSCCQPSCCETSSCCQPSCYQT-----SSCGTGCG
hKAP1.7	-----

repetitive I domain

hKAP1.1A/B	IGGGIGYGQEGSSGAVSTRIRWCRPDCRVEGTCLPPCCVVSCTPPSCCQLHHA
hKAP1.2	IGGGIGYGQEGSSGAVSTRIRWCRPDCRVEGTCLPPCCVVSCHTFTCCQLHHA
hKAP1.3	IGGGIGYGQEGSSGAVSTRIRWCRPDCRVEGTCLPPCCVVSCTPPTCCQLHHA
hKAP1.4	IGGGISYGQEGSSGSVSTRIRWCHPDCHVEGTCLPPCYLVSTPPSCCQLHHA
hKAP1.5	IGGGISYGQEGSSGAVSTRIRWCRPDSRVEGTCLPPCCVVSCTPPSCCQLHHA
hKAP1.6	IGGGIGYGQEGSSGAVSTRIRWCRPDCRVEGTCLPPCCVVSCTPPSCCQLHHA
hKAP1.7	-----SGAVSTRIRWCRPDCRVEGTCLPPCCVVSCTPPSCCQLHHA

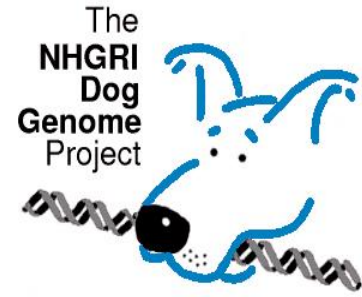
central non-repetitive domain

hKAP1.1A/B	EASCCRPSYCGQSCCRPVCCCYCSEPTC
hKAP1.2	EASCCRPSYCGQSCCRPVCCCYSCSHC-
hKAP1.3	EASCCRPSYCGQSCCRPVCCCYSCSEPTC
hKAP1.4	EASCCRPSYCGQSCCRPVCCCHCSEPTC
hKAP1.5	QASCCRPSYCGQSCCRPVCCCYCSEPTC
hKAP1.6	EASCCRPSYCGQSCCRPVCCCYCSEPTC
hKAP1.7	EASCCRPSYCGQSCCRPVCCCYCSEPTC

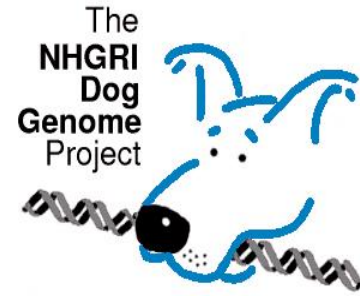
repetitive II domain

C-terminal domain

The Dog Genome Project: comparing genomes from different dog breeds to understand the genetic basis for their differences



The Dog Genome Project: comparing genomes from different dog breeds to understand the genetic basis for their differences



The NHGRI Dog Genome Project

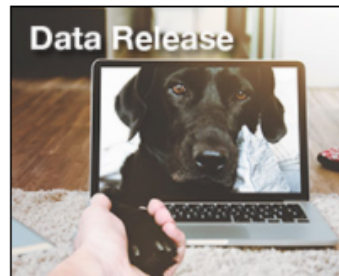
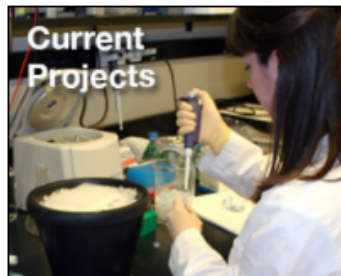
The NHGRI Dog Genome Project is run by Elaine Ostrander's Laboratory at NIH and focuses on the genetics of health and body structure in the domestic dog. We are particularly interested in the variation represented in individual dog breeds, which allows us to focus on locating genes involved in both canine cancer and the morphologic traits. Our research yields not only an understanding of how genes interact to create the modern dog breeds we see today and how we can keep our four-legged friends happy and healthy, but also insight into the health and disease of their two-legged caregivers.

Because of the breed structure, dogs display extensive variation in nearly all traits from size to coat color to behavior. There are giant and toy breeds; breeds with curly coats and those with straight coats; breeds with short snouts and those with longer snouts. Did you know overall body size varies forty-fold between large and little breeds? Modern dog breeds offer a great opportunity to understand the genetic basis of body plans and size in mammals.

In addition to differences in physical traits, breeds vary in disease risk. More than 350 inherited diseases have been described in domestic dogs, many of which predominate in a single breed or a small group of breeds. The NHGRI Dog Genome Project is working to find DNA variants that are associated with disease to help reduce their occurrence in affected breeds. Since many diseases in dogs are similar to disorders in humans, our findings will likely provide insight and new avenues of research for human health as well.

Would you like to interact with researchers at the NHGRI Dog Genome Project? We are always interested in the voluntary submission of DNA samples from all types of dogs as we have multiple studies underway at all times. We suggest you start by clicking through the material on this site and visiting our Facebook page.

You also can see our most recent publications, learn about events and/or dog shows we're attending, see research study plans, or ask questions to the experts in the lab.



Since humans and dogs share many similar genes and disorders, this information will inform research into human health as well.

Insights from comparing genomes within a species



14 different genes affect dog size

Breeds with short legs have an extra copy of the *FGF4* gene.

Some dog breeds have a high incidence of rare cancers. Why?

At least three genes control the length, texture and curl of a dog's coat.

[A locus linked to sociability in dogs \(and lacking in wolves\) corresponds to the human chromosomal deletion associated with Williams Syndrome in humans, which is associated with hyper-sociability.](#)

Comparing human genomes

This Genetic Mutation Makes People Feel Full — All the Time

Mutations in the MC4R gene usually lead to obesity by preventing a sense of fullness. But one such mutation leaves people uninterested in eating, scientists report.

The study subjects had been thin all their lives, and not because they had unusual metabolisms. They just did not care much about food.

Now, a group of researchers in Britain may have found the reason. The people carry a genetic alteration that mutes appetite. It also greatly reduces their chances of getting diabetes or heart disease. The scientists' study, published on Thursday in the journal Cell, relied on data from the U.K. Biobank, which includes a half million people aged 40 to 69. Participants have provided DNA samples and medical records, and have allowed researchers to track their health over years.

Read the news story at this [link](#).

Searching for the Genetic Underpinnings of Morning Persons and Night Owls

Researchers identified hundreds of gene variations that may be connected to when people go to bed.

The researchers drew on data from 23andMe, the genetic testing company, and the UK Biobank, which tracks hundreds of thousands of volunteer subjects in Britain, about 85,000 of whom wear activity monitors that record their movements.

Read the news story at this [link](#).

Their DNA Hides a Warning, but They Don't Want to Know What It Says

Some volunteers for biobanks, which collect genetic information to study health across populations, have been surprised to be informed they carry risky mutations.

Read the news story at this [link](#).

Comparative genomics in the literature

Research article excerpt

Comparative Genomics and Human Medicine

Sequence comparisons within large databases

Characterization of Prevalence and Health Consequences of Uniparental Disomy in Four Million Individuals from the General Population

The American Journal of Human Genetics

Volume 105, Issue 5, Pages 921-932 (November 2019)

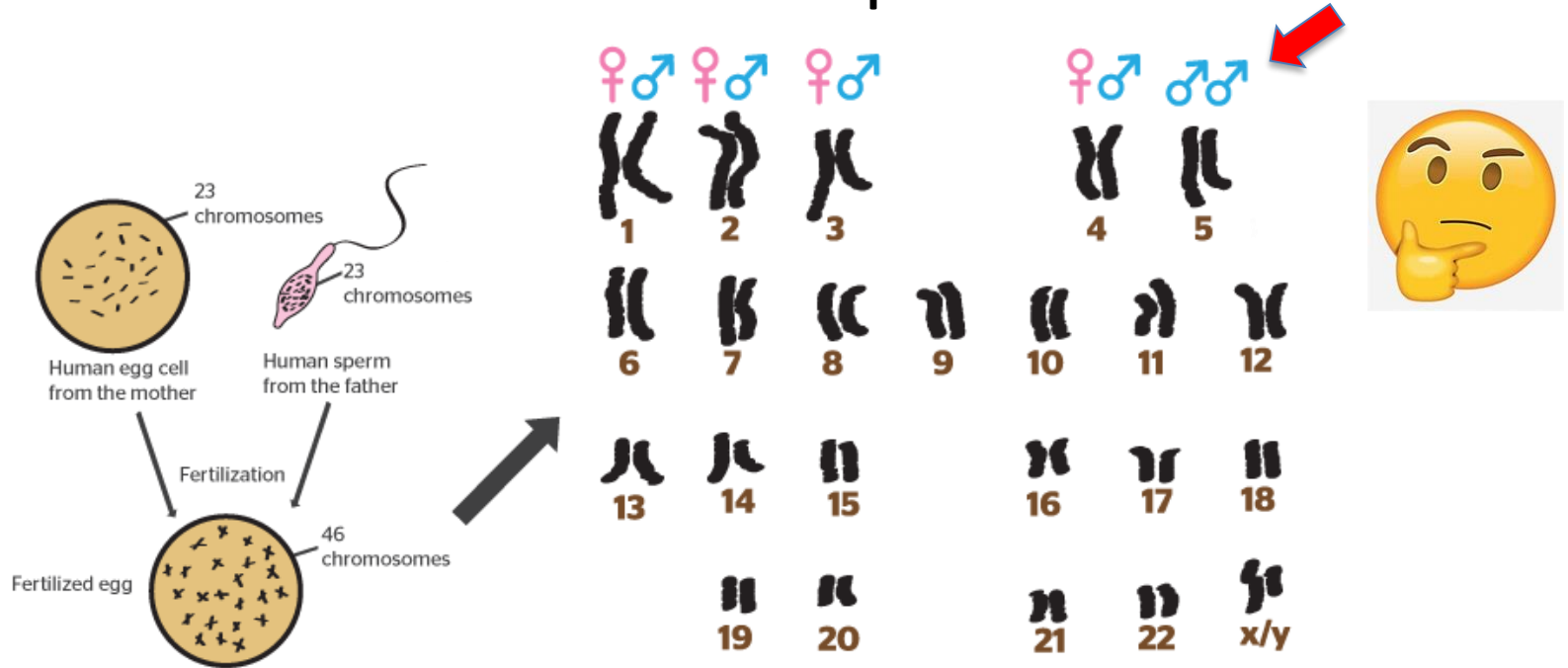
DOI: 10.1016/j.ajhg.2019.09.016

Unbalanced Chromosomal Inheritance More Common than Thought

A genomic analysis from 23andMe suggests that people inherit two copies of a chromosome from only one parent nearly twice as often as researchers had realized.

<https://www.the-scientist.com/news-opinion/unbalanced-chromosomal-inheritance-more-common-than-thought-66563>

A child normally inherits one copy of each chromosome from each parent. But....

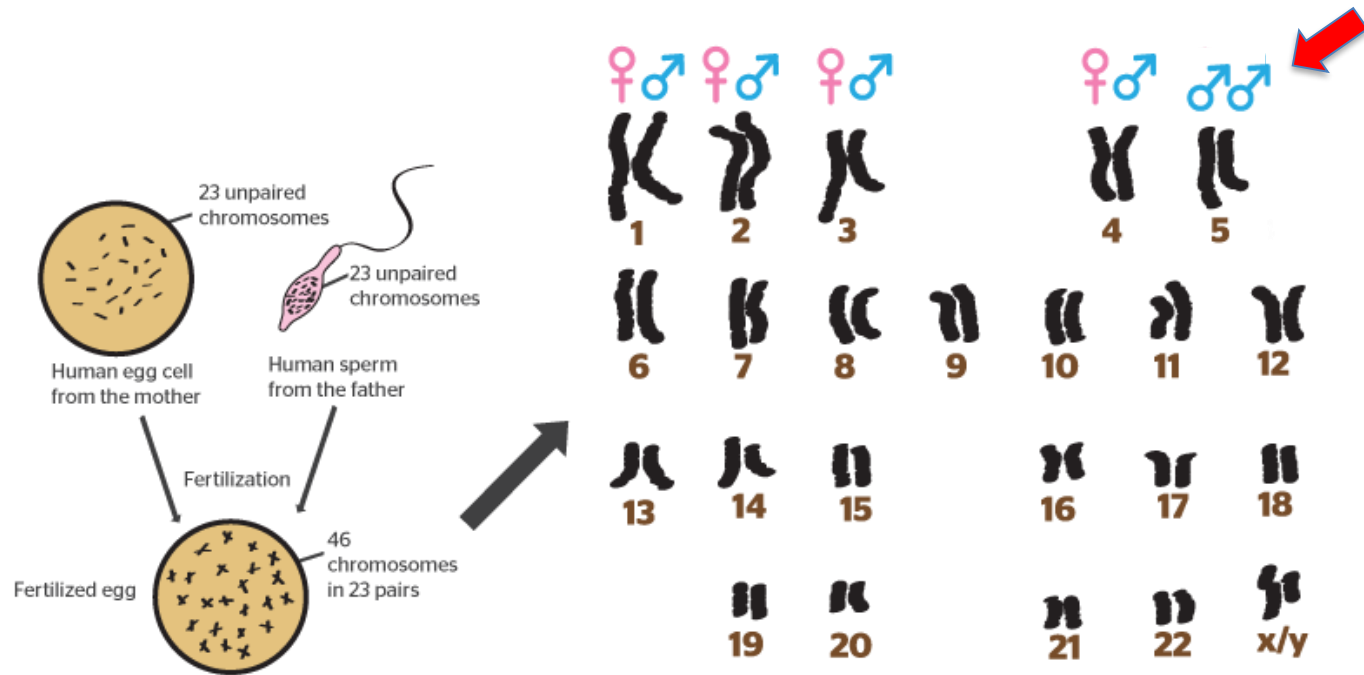


Identified 916,712 “Parent-Child Duos,” i.e. cases where they could tell that one individual was the parent of another individual

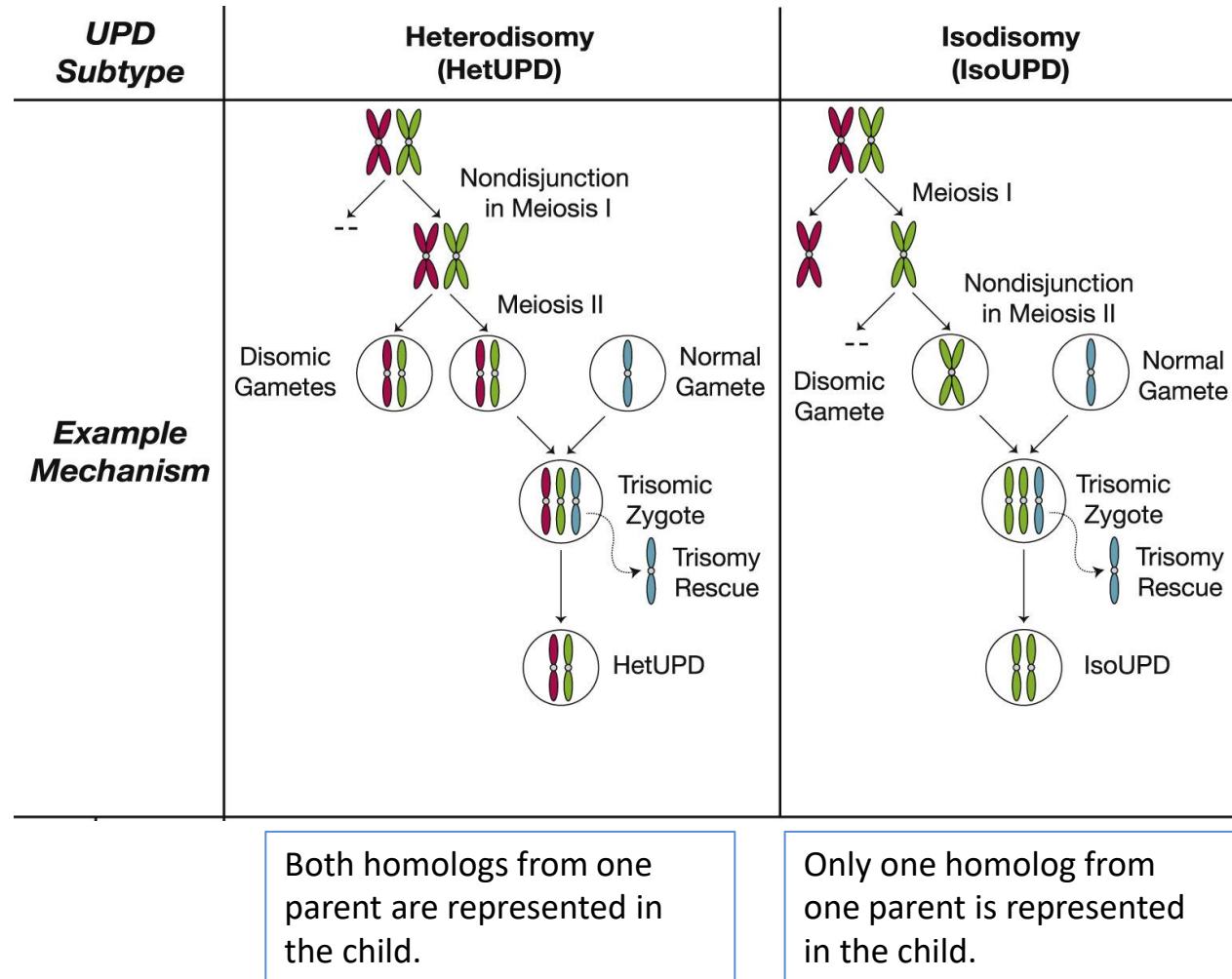
Of these 916,712, they found 199 individuals that have two copies of a chromosome from one parent and no copies of that chromosome from the other parent.

This is called: Uniparental Disomy

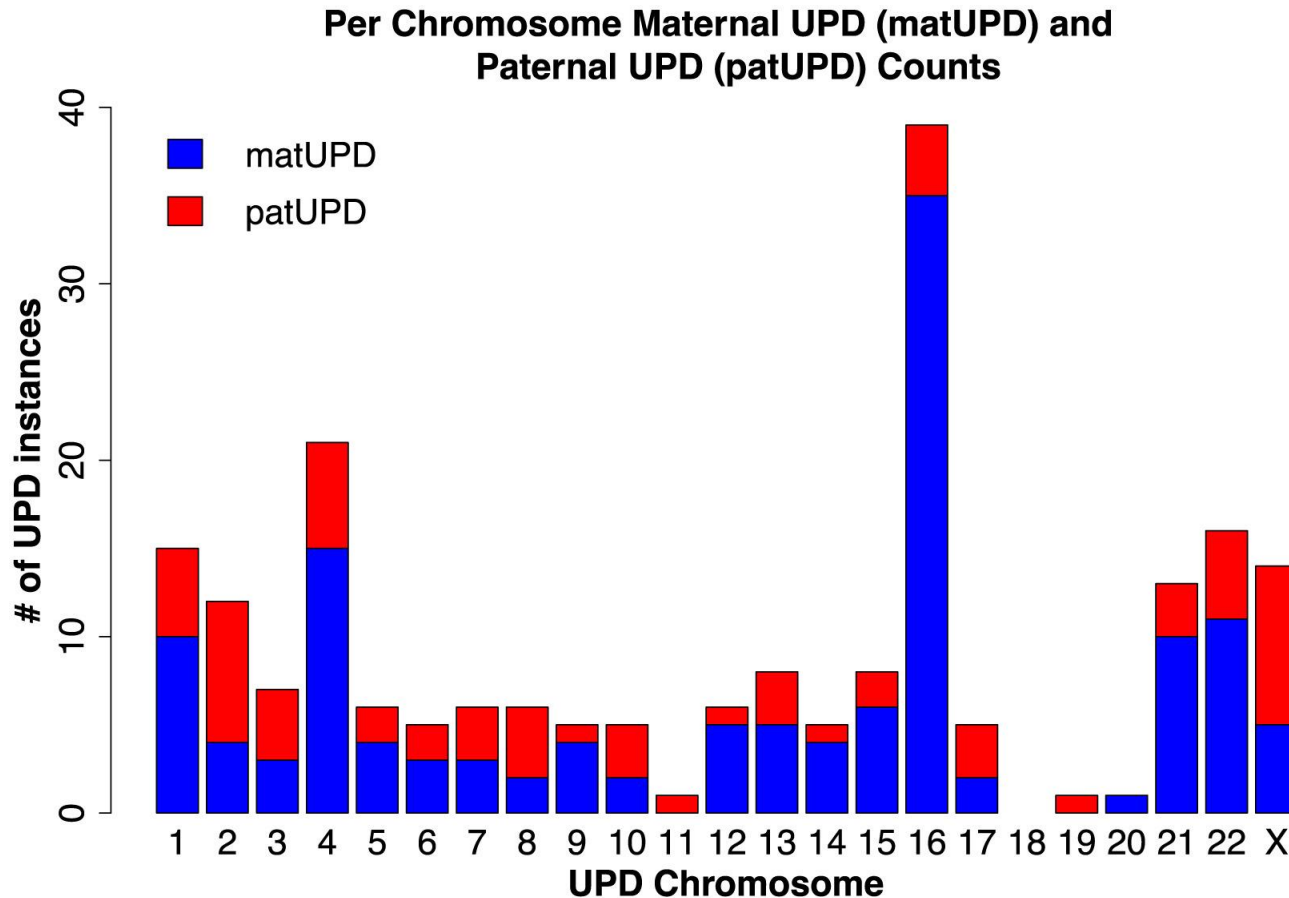
How can two chromosomes from the same parent end up in the same F1 individual?



How can two chromosomes from the same parent end up in the same F1 individual?



Not all chromosomes are equally likely to be found in UPD



“Using IBD-Based UPD Detection, we found 199 individuals with 205 incidences of UPD in 916,712 Parent-Child Duos from the 23andMe dataset (.005%, or about 1 in 20,000).”